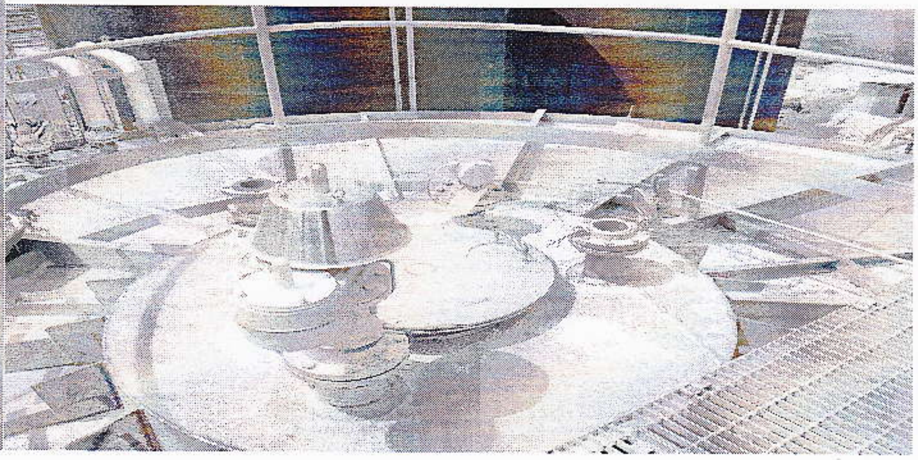
Products Solutions Services

Training Level Radar Basic Programing

Wilmar Indonesia
Gresik



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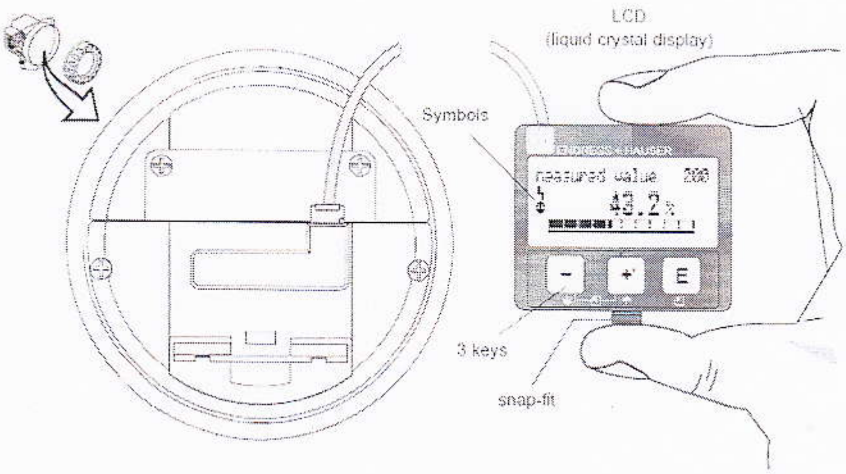
Operation Display and operating elements



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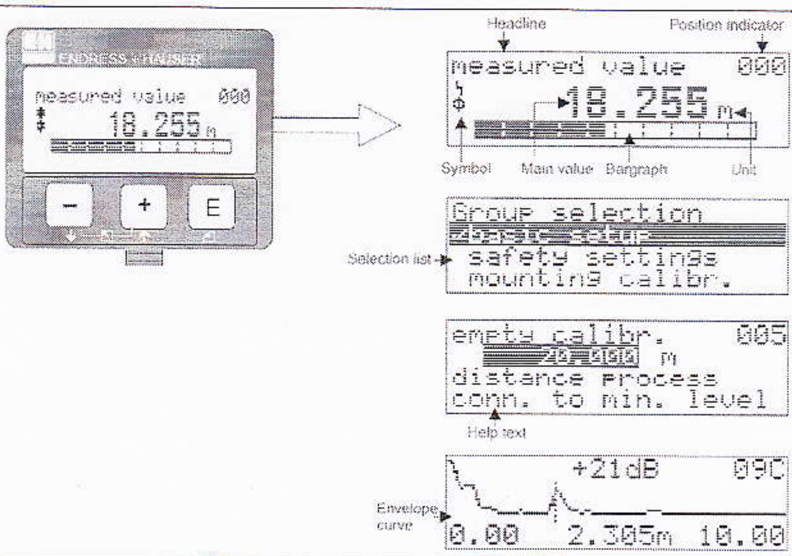
Display



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Display symbol

Symbol	Meaning
	ALARM_SYMBOL This alarm symbol appears when the instrument is in an alarm state. If the symbol flashes, this indicates a warning.
	LOCK_SYMBOL This lock symbol appears when the instrument is locked, i.e. if no input is possible.
	COM_SYMBOL This communication symbol appears when a data transmission via e.g. HART, PROFIBUS PA or FOUNDATION Fieldbus is in progress.

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Key assignment

Key(s)	Meaning
+ or ↑	Navigate upwards in the selection list. Edit numeric value within a function.
− or ↓	Navigate downwards in the selection list. Edit numeric value within a function.
− + or ←	Navigate to the left within a function group.
E	Navigate to the right within a function group, confirmation.
+ and E or − and E	Contrast settings of the LCD.
+ and − and E	Hardware lock / unlock After a hardware lock, an operation of the instrument via display or communication is not possible! The hardware can only be unlocked via the display. An unlock parameter must be entered to do so.

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Overview Basic Setup

level measurement in liquids

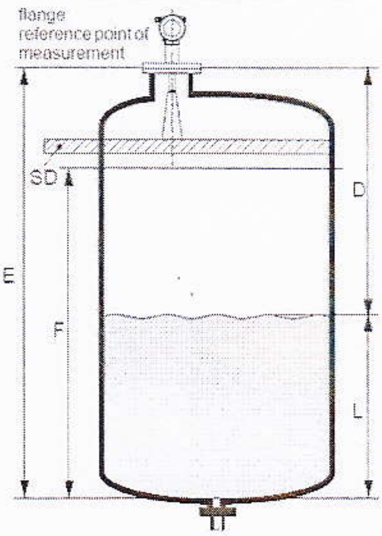
E = empty calibr. (= zero), setting in 005

F = full calibr. (= span), setting in 006

D = distance (distance flange / product), display in 0A5

L = level, display in 0A6

SD = safety distance, setting in 015



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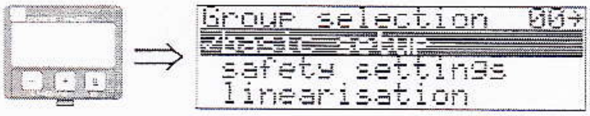
Basic Setup with the device display

Measured value

Basic setup



Media type

Media type

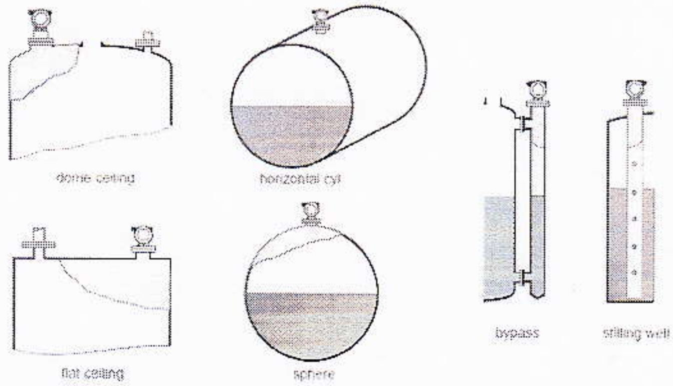
With the selection "liquid" only the following functions can be adjusted:

- tank shape 002
- medium property 003
- process cond. 004
- empty calibr. 005
- full calibr. 006
- pipe diameter 007
- check distance 051
- range of mapping 052
- start mapping 053
- ...

With the selection "solids" only the following functions can be adjusted:

- vessel / silo 00A
- medium property 00B
- process cond. 00C
- empty calibr. 005
- full calibr. 006
- check distance 051
- range of mapping 052
- start mapping 053
- ....

Tank Shape



Medium Property

- Liquid

Media group	DC (εr)	Examples
A	< 1.9	non-conducting liquids, e.g. liquefied gas[1]
B	1.9 to 4	non-conducting liquids, e.g. benzene, oil, toluene, ...
C	4 to 10	e.g. concentrated acids, organic solvents, esters, aniline, alcohol, acetone, ...
D	> 10	conducting liquids, e.g. aqueous solutions, dilute acids and alkalis

- Solid

Media group	DC (εr)	Examples
A	1.6 to 1.9	Plastic granulate, White lime, special cement, sugar
B	1.9 to 2.5	Portland cement, plaster
C	2.5 to 4	Grain, seeds, ground stones, sand
D	4 to 7	naturally moist (ground) stones, ores, salt
E	> 7	Metallic powder, carbon black, coal

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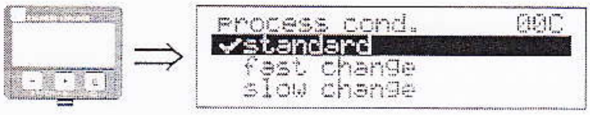
Process Condition

Selection Liquid:

- **standard**
- calm surface
- turb. surface
- agitator
- fast change
- test: no filter

Selection Solid:

- **standard**
- fast change
- slow change
- test: no filter



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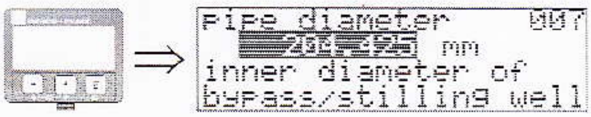
Basic Function

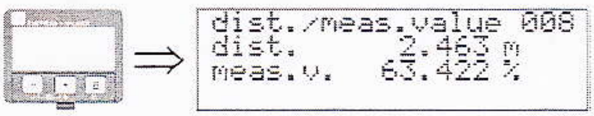
Empty Calibration

Full Calibration

Pipe diameter (only for bypass/ Stillingwell)



Dist./meas value



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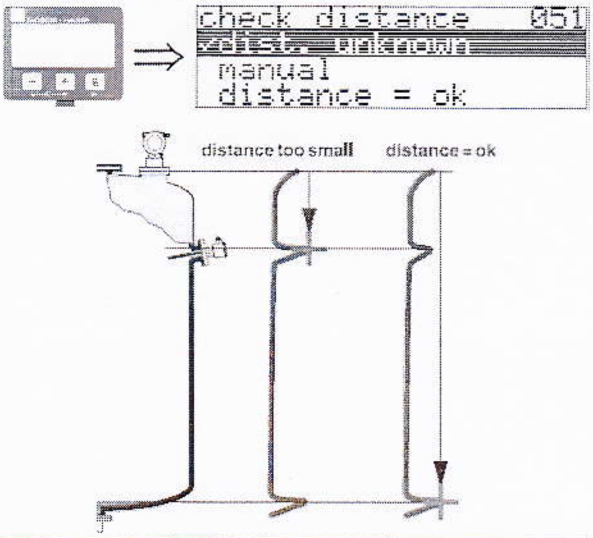
Check Distance

Selection:

- distance = ok
- dist. too small
- dist. too big
- **dist. unknownt**
- manual



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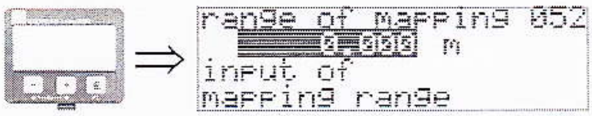
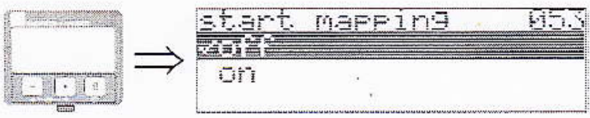
Mapping

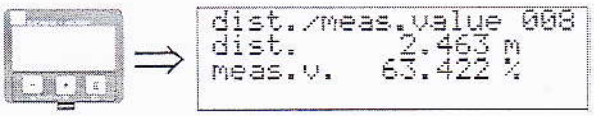
Range of mapping



Start mapping



Distance Measure value



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Final Check

The **distance** measured from the reference point to the product surface and the **level** calculated with the aid of the empty adjustment are displayed. Check whether the values correspond to the actual meas. value or the actual distance. The following cases can occur:

- Distance correct - level correct È continue with the next function, "check distance" (051)
- Distance correct - level incorrect È Check "empty calibr." (005)
- Distance incorrect - level incorrect È continue with the next function "check distance" (051).

⇒

Return to
Group Selection

⇩

Group selection 00→
basic setup
safety settings
length adjustment

Training Level Radar

Any questions?